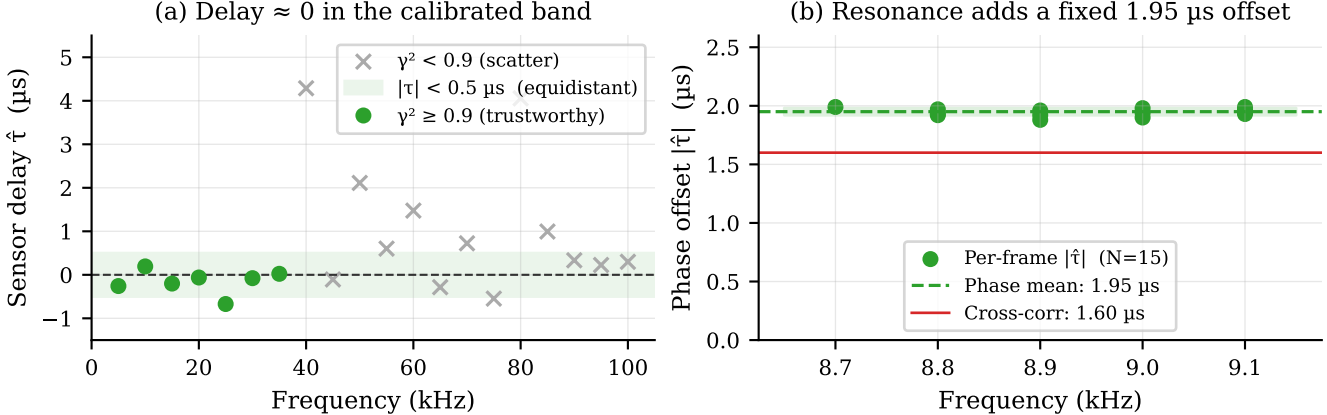
(a) Delay ≈ 0 in the calibrated band
Sensor delay τ̂ (µs)
Frequency (kHz)
γ² < 0.9 (scatter)
|τ| < 0.5 µs (equidistant)
γ² ≥ 0.9 (trustworthy)
(b) Resonance adds a fixed 1.95 µs offset
Phase offset |τ̂| (µs)
Frequency (kHz)
Per-frame |τ̂| (N=15)
Phase mean: 1.95 µs
Cross-corr: 1.60 µs